

Why No Existing Mathematical Theory of Evolution Includes Non-Existence Domains

Abstract

Since the early twentieth century, several foundational figures—Fisher, Wright, Haldane, Eigen, and later Kauffman—have attempted to formalize biological evolution using mathematical models. These efforts successfully quantified rates, equilibria, stochastic effects, and information limits within evolutionary dynamics. However, all existing mathematical theories of evolution share a critical and largely unexamined assumption: that a viable evolutionary state space always exists. This paper argues that none of these frameworks incorporates *non-existence domains*—regions of boundary conditions under which no coherent biological configuration is possible. We show that this omission is not technical but conceptual, and that it fundamentally limits the explanatory power of Darwinian and post-Darwinian models in regimes involving abrupt environmental transitions, mass extinctions, and macroevolutionary phase shifts.

1. Introduction: Formal Evolution Without Failure States

Mathematical theories of evolution were developed to replace narrative biology with predictive structure. From population genetics to information-theoretic models, evolution has been treated as a dynamical process evolving over time within a defined state space.

Yet a striking commonality persists:

all formal evolutionary models presuppose that a solution exists.

Fitness landscapes may be rugged, mutation rates may be high, and dynamics may be noisy or chaotic—but the system is always assumed to remain within a space of viable biological states.

This paper examines why no existing mathematical theory of evolution allows for the possibility that *no evolutionary trajectory exists at all* under certain boundary conditions.

2. Fisher: Evolution as Continuous Improvement

Ronald Fisher's *Fundamental Theorem of Natural Selection* formalized evolution as a law-like increase in mean fitness proportional to genetic variance. This was a decisive step toward treating evolution as a physical process.

However, Fisher's framework assumes:

- continuous time,
- continuous variation,
- and a well-defined fitness function at all points.

Fitness may increase slowly or rapidly, but it is never undefined. The theorem cannot represent situations where **no phenotype satisfies the environmental constraints**. In Fisher's model, evolution may slow—but it cannot fail to exist.

3. Wright: Rugged Landscapes Without Void Regions

Sewall Wright introduced adaptive landscapes with multiple peaks and valleys, capturing local optima, genetic drift, and population structure. His work acknowledged topological complexity and nonlinearity.

Yet even Wright's landscapes are *total*: every point corresponds to a viable (if suboptimal) genotype.

Valleys may be deep, but they are traversable in principle. The landscape never collapses. There are no regions where fitness is undefined or biologically meaningless. Thus, Wright allows difficulty—but not impossibility.

4. Haldane: Rates and Costs Without Structural Breakdown

J.B.S. Haldane confronted evolutionary speed directly, calculating substitution costs and limits on how fast populations can change. His work comes closest to exposing contradictions in gradualism.

However, when calculated costs exceed plausible biological limits, the conclusion is never structural failure. Instead, the explanation is deferred:

- longer timescales,

- larger populations,
- unobserved mechanisms.

The model is stretched, but never abandoned. The possibility that the evolutionary framework itself ceases to apply is never entertained.

5. Eigen: Error Thresholds Without Architectural Transitions

Manfred Eigen's quasispecies model introduced a genuine information-theoretic constraint: the **error threshold**. Above a critical mutation rate, genetic information collapses.

This is a profound result: evolution becomes computationally impossible beyond a limit.

Yet even here, the state space remains fixed. The genome exists; replication exists; fitness exists. The model predicts collapse, but not **replacement by a new structural regime**. What lies beyond the threshold is left formally undefined.

6. Kauffman: Self-Organization Without Non-Existence

Stuart Kauffman proposed that life emerges naturally in complex systems through self-organization and criticality. His models emphasize robustness and inevitability.

But inevitability is precisely the limitation. In Kauffman's framework, order always emerges somewhere. The system reorganizes, adapts, or stabilizes—but never confronts a boundary where *no coherent biological organization is possible*.

Failure is aesthetic, not ontological.

7. The Missing Concept: Non-Existence Domains

Across all these theories, a shared assumption persists:

For any set of environmental conditions, there exists at least one biologically viable configuration.

This assumption is never proven; it is inherited.

A **non-existence domain** is a region of boundary conditions in which:

- no phenotype is viable,
- no adaptive trajectory exists,
- and no gradual path connects past and future biological architectures.

Such domains imply that evolution, as traditionally modeled, **cannot operate**. What follows is not adaptation, but **phase transition**—the collapse and reconstitution of biological structure at a higher organizational level.

8. Implications for Macroevolution and Extinction Events

Phenomena such as:

- the Cambrian explosion,
- mass extinctions,
- rapid architectural innovation,
- and abrupt phenotypic expansion,

cannot be adequately modeled within frameworks that forbid non-existence domains by assumption.

Trying to explain such events using fossil intermediates alone is analogous to arranging a row of candles and expecting them to generate a lightning bolt. The phenomenon is real, but the explanatory scale is wrong.

9. Conclusion

No existing mathematical theory of evolution includes non-existence domains—not because they are mathematically intractable, but because they violate a deep conceptual commitment to continuity.

Darwinian evolution, population genetics, and even information-based models describe **local dynamics within stable regimes**. They do not—and cannot—account for situations where the evolutionary state space itself disappears.

Recognizing this limitation does not falsify evolution.
It **defines its domain of validity**.

Beyond that domain, a different kind of theory is required—one that treats life not as a perpetual optimizer, but as a system constrained by minimal coherence, boundary conditions, and phase transitions.

Appendix — Why Continuity Is an Unproven Assumption in Evolutionary Theory

A.1 The Hidden Axiom of Continuity

All classical and modern evolutionary theories implicitly assume that evolutionary change is continuous—or at least piecewise continuous—over time. This assumption is rarely stated explicitly, yet it underlies:

- the use of differential equations in population genetics,
- continuous fitness landscapes,
- gradual allele frequency shifts,
- and the notion of evolutionary trajectories connecting past and future phenotypes.

Formally, this corresponds to assuming that for any admissible change in boundary conditions, there exists a nearby viable biological state. This is an **axiom**, not a derived result.

A.2 Continuity Is Not Entailed by Darwinian Logic

Darwinian evolution requires only three elements:

1. variation,
2. inheritance,
3. differential survival or reproduction.

None of these logically implies continuity of viable phenotypes under changing conditions.

Selection can operate only if:

- viable variants already exist,
- and the fitness function is well-defined.

If boundary conditions shift beyond a critical threshold, selection does not “slow down” or “struggle”; it simply becomes undefined. Continuity is therefore an *extra assumption*, not a consequence of Darwinian principles.

A.3 Mathematical Convenience vs. Ontological Necessity

Continuity is mathematically convenient. It allows:

- linearization,
- gradient ascent,
- smooth optimization,
- and tractable analytical models.

However, mathematical convenience does not imply ontological truth.

In physics, continuity was long assumed until quantum mechanics demonstrated discrete state spaces and forbidden regions. No analogous justification exists for assuming that biological viability is continuous across all environmental regimes.

A.4 Fossil Continuity Is an Observational Bias

The apparent continuity of the fossil record is often cited as empirical support for gradualism. However:

- fossils sample survivable states only,
- non-viable configurations leave no record,
- and extinction events erase entire regions of phenotype space.

Thus, the fossil record cannot falsify the existence of non-existence domains; it is blind to them by construction.

A.5 Error Thresholds Already Violate Continuity

Information-theoretic models (e.g., Eigen’s error threshold) demonstrate that beyond a critical mutation rate, heredity collapses.

This introduces a **hard boundary**:

- below the threshold, evolution operates;
- above it, genetic information cannot be maintained.

Yet most evolutionary frameworks treat this as a pathological edge case rather than acknowledging it as evidence that continuity is not guaranteed.

A.6 Boundary Conditions Can Eliminate the State Space

Let B denote environmental boundary conditions and $\Omega(B)$ the set of viable biological configurations.

Classical evolutionary theory assumes:

$$\forall B, \quad \Omega(B) \neq \emptyset$$

This assumption is never justified.

A regime transition occurs when:

$$\Omega(B) = \emptyset$$

At this point:

- selection cannot act,
- gradual trajectories cease,
- and continuity fails.

What follows is not evolution in the Darwinian sense, but a **structural phase transition**.

A.7 Why the Assumption Persisted

The continuity assumption persisted historically because:

- early evolutionary change appeared slow at human timescales,
- available data was sparse and coarse-grained,
- and catastrophic transitions were underrepresented or misinterpreted.

As a result, gradualism became a default framework rather than a tested hypothesis.

A.8 Implication: Evolution Is Locally Valid, Not Globally Defined

The correct conclusion is not that evolution is false, but that it is **locally valid** within regimes where:

- viable configurations exist,
- mutation rates remain below structural thresholds,
- and boundary conditions change slowly relative to biological response times.

Outside these regimes, continuity is not preserved, and evolutionary theory must be supplemented by a framework that explicitly models non-existence domains and phase transitions.

Closing Remark

Continuity in evolution is not an empirical discovery.

It is an assumption inherited from mathematical convenience and historical context.

Recognizing its unproven status is not a rejection of evolutionary theory, but a necessary step toward defining its true domain of applicability.